LOOP CHECK

Operator’s Guide

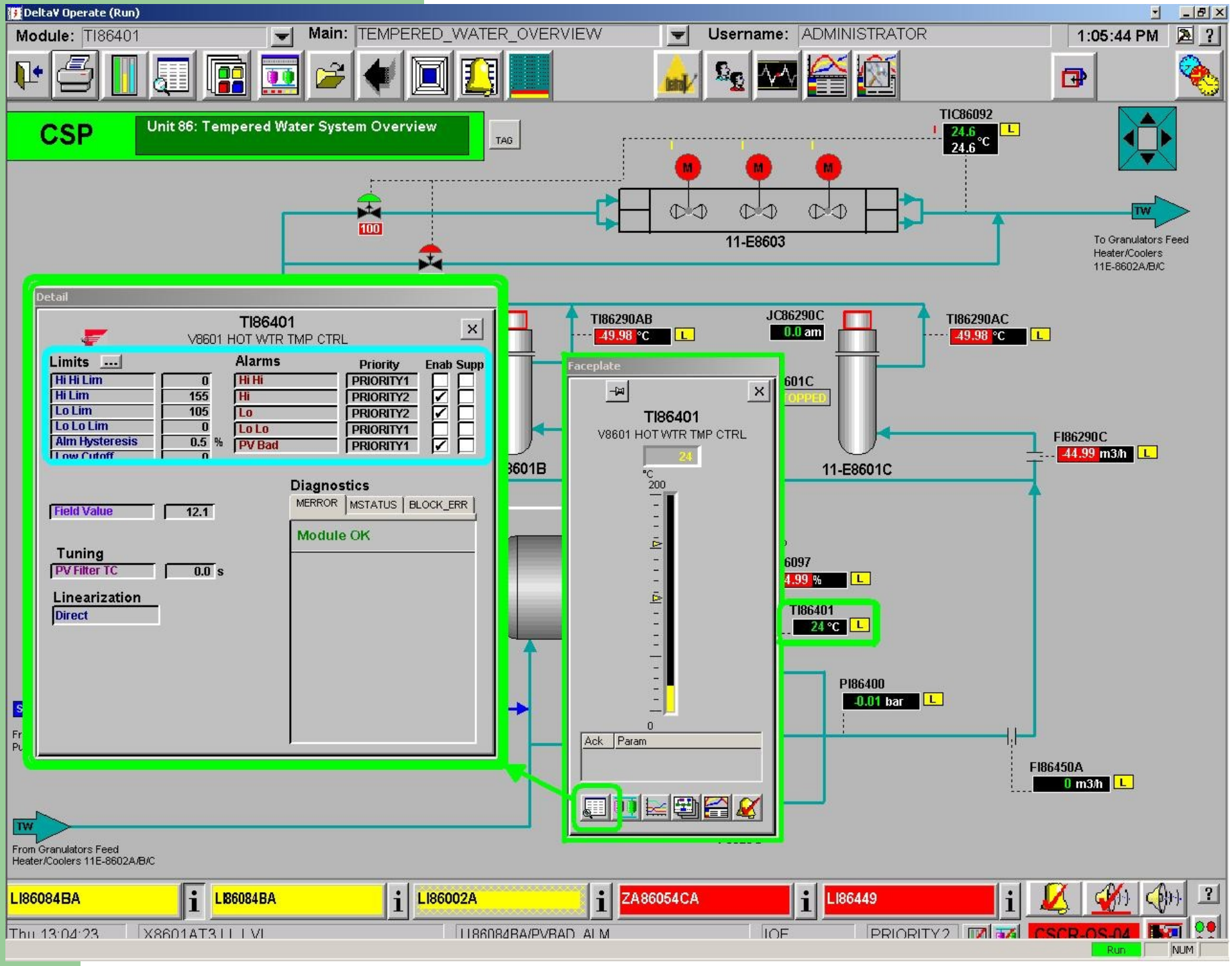
4-20mA signals (Conventional, HART)

- Verify that the applicable parameters (span, alarm set point) are correct (Loop folder INTools data as the official reference)

4-20mA signals (Conventional, HART)

Analog Input

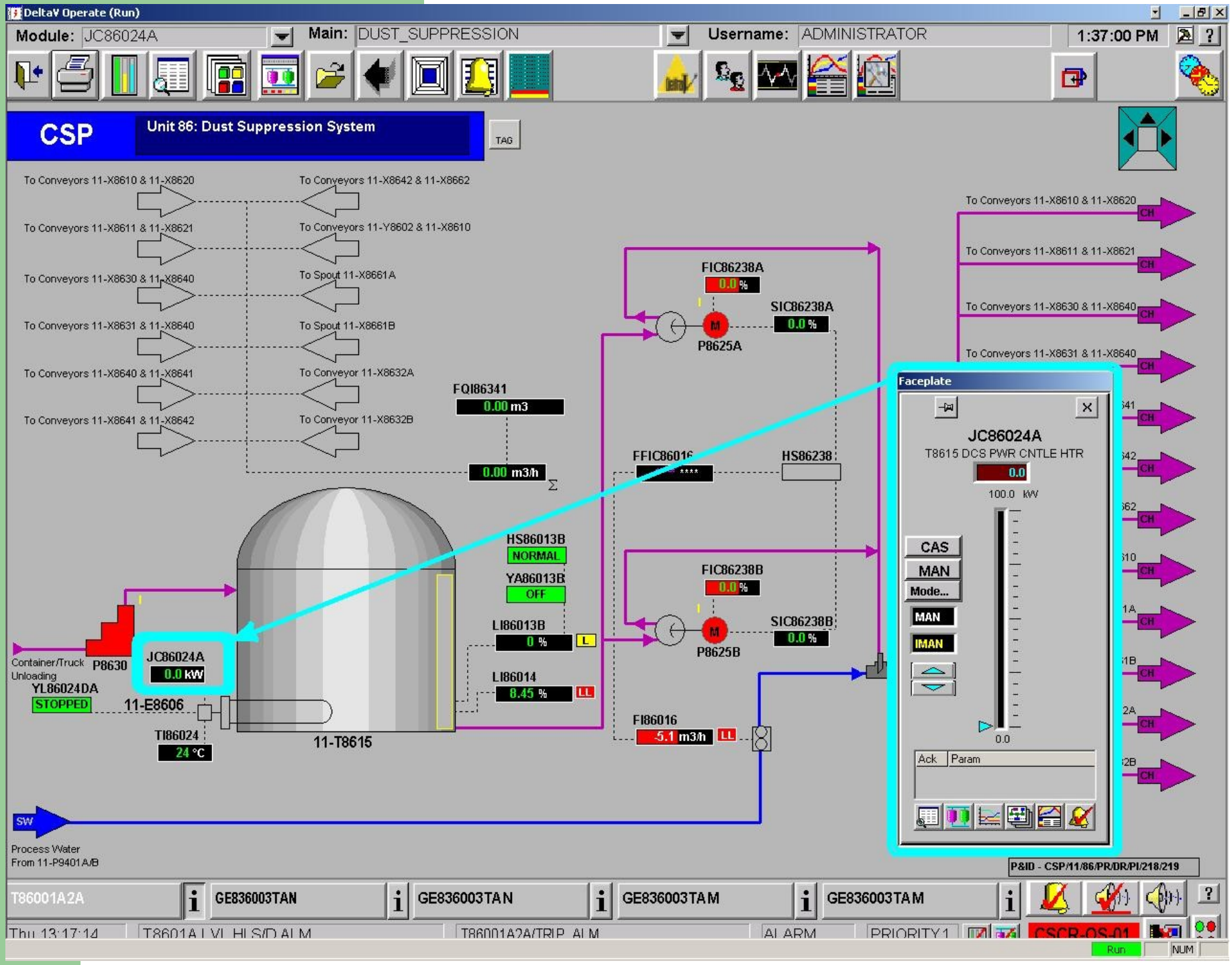
- Field:
  - force input to 0%, 50%, 100% of span
- Control Station:
  - verify correct reading in DeltaV operate (Control graphics / Faceplate)
  - verify alarm (HH, Hi, Lo, LL) Settings (Detailed Faceplate)
  - Check burnout for Thermocouple loop


DeltaV Operate (Run)
Module: TI86401
Main: TEMPERED_WATER_OVERVIEW
Username: ADMINISTRATOR
1:05:44 PM
CSP
Unit 86: Tempered Water System Overview
TAG
TIC86092
24.6
24.6 °C
11-E8603
To Granulators Feed
Heater/Coolers
11E-8602A/B/C
TI86290AB
-49.98 °C
JC86290C
0.0 am
TI86290AC
-49.98 °C
FI86290C
-44.99 m3/h
11-E8601C
Detail
TI86401
V8601 HOT WTR TMP CTRL
Limits
Alarms
Priority
Enab
Supp
Hi Hi Lim 0 Hi Hi PRIORITY1
Hi Lim 155 Hi PRIORITY2
Lo Lim 105 Lo PRIORITY2
Lo Lo Lim 0 Lo Lo PRIORITY1
Alm Hysteresis 0.5 % PV Bad PRIORITY1
Low Cutoff 0
Diagnostics
MERROR
MSTATUS
BLOCK_ERR
Module OK
Field Value 12.1
Tuning
PV Filter TC 0.0 s
Linearization
Direct
Faceplate
TI86401
V8601 HOT WTR TMP CTRL
24
°C
200
0
Ack
Param
TI86401
24 °C
PI86400
-0.01 bar
FI86450A
0 m3/h
From Granulators Feed
Heater/Coolers 11E-8602A/B/C
LI86084BA
LI86084BA
LI86002A
ZA86054CA
LI86449
Thu 13:04:23
Run
NUM

4-20mA signals (Conventional, HART)

Analog Output

- Control Station:
  - Open Control Faceplate.
  - Force output to 0%, 50%, 100% of span
- Field:
  - verify that 4, 12, 20 mA is received.
  - verify device action if applicable


DeltaV Operate (Run)
Module: JC86024A
Main: DUST_SUPPRESSION
Username: ADMINISTRATOR
1:37:00 PM
CSP
Unit 86: Dust Suppression System
TAG
To Conveyors 11-X8610 & 11-X8620
To Conveyors 11-X8642 & 11-X8662
To Conveyors 11-X8611 & 11-X8621
To Conveyors 11-Y8602 & 11-X8610
To Conveyors 11-X8630 & 11-X8640
To Spout 11-X8661A
To Conveyors 11-X8631 & 11-X8640
To Spout 11-X8661B
To Conveyors 11-X8640 & 11-X8641
To Conveyor 11-X8632A
To Conveyors 11-X8641 & 11-X8642
To Conveyor 11-X8632B
FQI86341
0.00 m3
0.00 m3/h
FIC86238A
0.0 %
SIC86238A
0.0 %
P8625A
FFIC86016
HS86238
HS86013B
NORMAL
YA86013B
OFF
FIC86238B
0.0 %
SIC86238B
0.0 %
P8625B
LI86013B
0 %
L
LI86014
8.45 %
LL
FI86016
-5.1 m3/h
LL
Container/Truck Unloading
P8630
JC86024A
0.0 kW
YL86024DA
STOPPED
11-E8606
TI86024
24 °C
11-T8615
SW
Process Water
From 11-P9401A/B
To Conveyors 11-X8610 & 11-X8620
To Conveyors 11-X8611 & 11-X8621
To Conveyors 11-X8630 & 11-X8640
To Conveyors 11-X8631 & 11-X8640
CH
Faceplate
JC86024A
T8615 DCS PWR CNTLE HTR
0.0
100.0 kW
CAS
MAN
Mode...
MAN
IMAN
0.0
Ack
Param
P&ID - CSP/11/86/PR/DR/PI/218/219
T86001A2A
GE836003TAN
GE836003TAN
GE836003TAM
GE836003TAM
Thu 13:17:14
T8601A LVL HI S/D ALM
T86001A2A/TRIP ALM
ALARM
PRIORITY 1
CSCR-OS-01
Run
NUM

Digital Input/Output signals

- Verify that the alarm set-point is correct (0 or 1 to alarm - Loop folder INTools data as the official reference)

Digital Input/Output signals

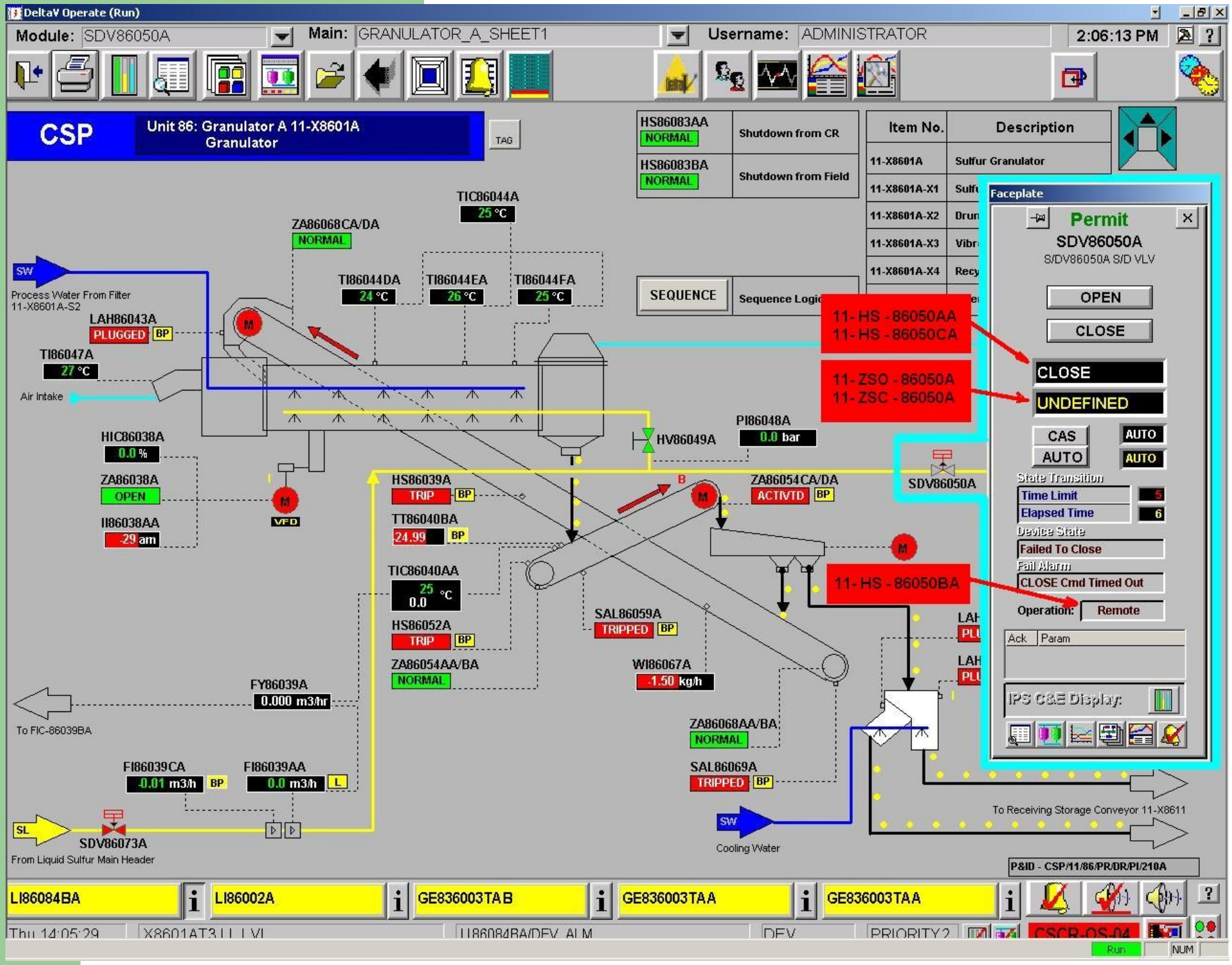
Digital input

- Filed:
  - activate field device if applicable.
  - (switches –level, pressure, temperature, flow, Remote/Local, Start/Stop, etc.)
- Control Station:
  - verify correct indication in DeltaV operate (Control graphics / Faceplate) if applicable.

Digital Input/Output signals

Digital output

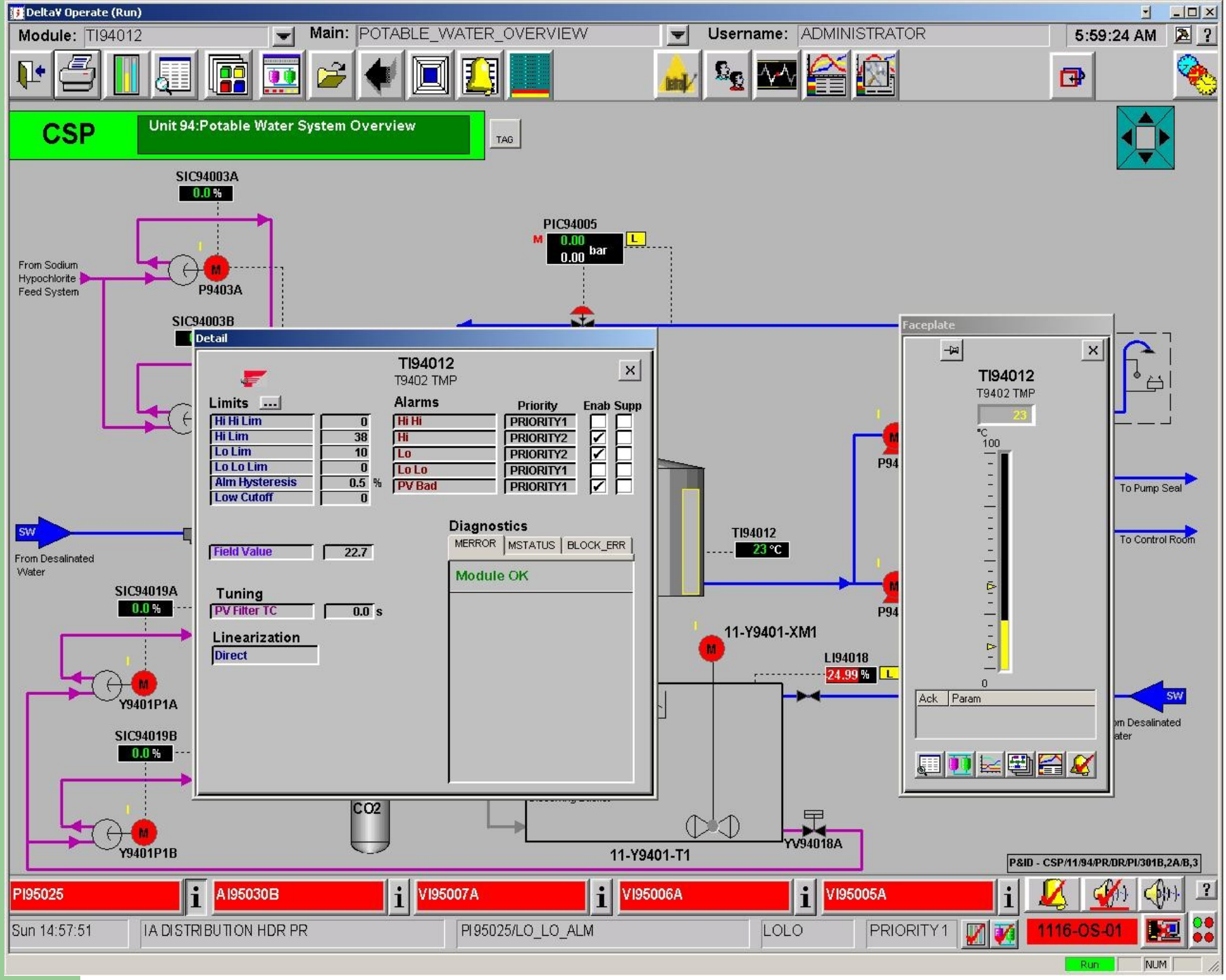
- Control Station:
  - Open Control Faceplate
  - Force output to energized\deenergized, open, close
- Field:
  - verify that Control voltage is received
  - verify device action if applicable


DeltaV Operate (Run)
Module: SDV86050A
Main: GRANULATOR_A_SHEET1
Username: ADMINISTRATOR
2:06:13 PM
CSP
Unit 86: Granulator A 11-X8601A
Granulator
TAG
HS86083AA
NORMAL
Shutdown from CR
HS86083BA
NORMAL
Shutdown from Field
Item No.
Description
11-X8601A
Sulfur Granulator
11-X8601A-X1
11-X8601A-X2
11-X8601A-X3
11-X8601A-X4
SEQUENCE
Sequence Logic
TIC86044A
25 °C
ZA86068CA/DA
NORMAL
TI86044DA
24 °C
TI86044EA
26 °C
TI86044FA
25 °C
SW
Process Water From Filter
11-X8601A-S2
LAH86043A
PLUGGED
BP
TI86047A
27 °C
Air Intake
HIC86038A
0.0 %
ZA86038A
OPEN
II86038AA
-29 am
VFD
HV86049A
PI86048A
0.0 bar
HS86039A
TRIP
BP
TT86040BA
24.99
BP
ZA86054CA/DA
ACTIVTD
BP
SDV86050A
TIC86040AA
25 °C
0.0
HS86052A
TRIP
BP
ZA86054AA/BA
NORMAL
SAL86059A
TRIPPED
BP
WI86067A
-1.50 kg/h
FY86039A
0.000 m3/hr
To FIC-86039BA
FI86039CA
-0.01 m3/h
BP
FI86039AA
0.0 m3/h
L
SL
SDV86073A
From Liquid Sulfur Main Header
ZA86068AA/BA
NORMAL
SAL86069A
TRIPPED
BP
SW
Cooling Water
To Receiving Storage Conveyor 11-X8611
P&ID - CSP/11/86/PR/DR/PI/210A
11-HS-86050AA
11-HS-86050CA
11-ZSO-86050A
11-ZSC-86050A
11-HS-86050BA
Faceplate
Permit
SDV86050A
S/DV86050A S/D VLV
OPEN
CLOSE
CLOSE
UNDEFINED
CAS
AUTO
AUTO
AUTO
State Transition
Time Limit
5
Elapsed Time
6
Device State
Failed To Close
Fail Alarm
CLOSE Cmd Timed Out
Operation:
Remote
Ack
Param
IPS C&E Display:
LI86084BA
LI86002A
GE836003TAB
GE836003TAA
GE836003TAA
Thu 14:05:29
X8601AT3LLLVL
LI86084BA/DEV ALM
DEV
PRIORITY2
CSCR-OS-04
Run
NUM

Fieldbus Devices

Transmitters (Pressure, Temperature, Level, Flow, etc)

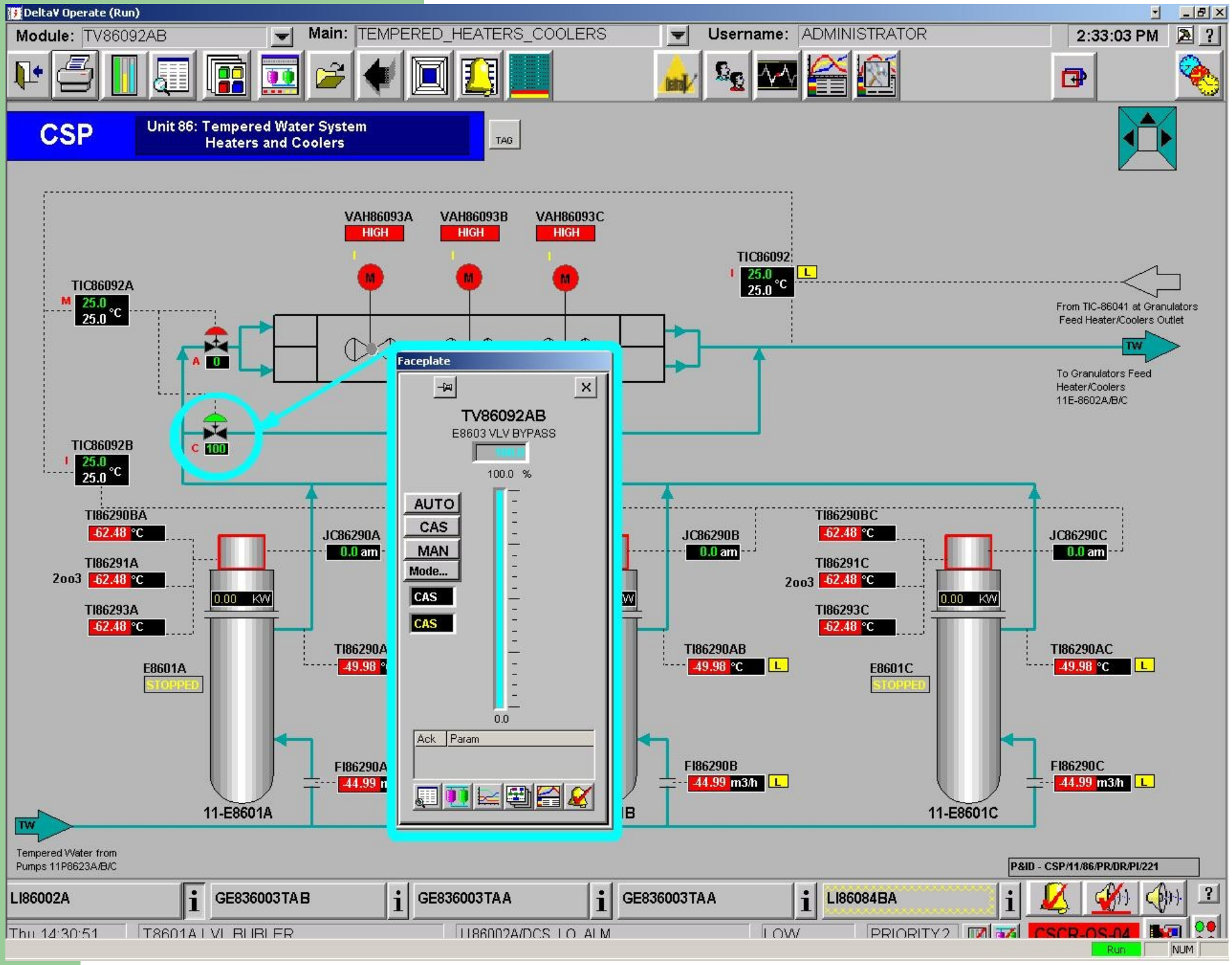
- Field:
  - force input to 0%, 50%, 100% of span (Inject signal – pressure, temperature, etc. whichever is applicable. Or simulate PV using AMS)
- Control Station:
  - verify correct reading in DeltaV operate (Control graphics / Faceplate)
  - verify alarm (HH, Hi, Lo, LL) Settings (Detailed Faceplate)

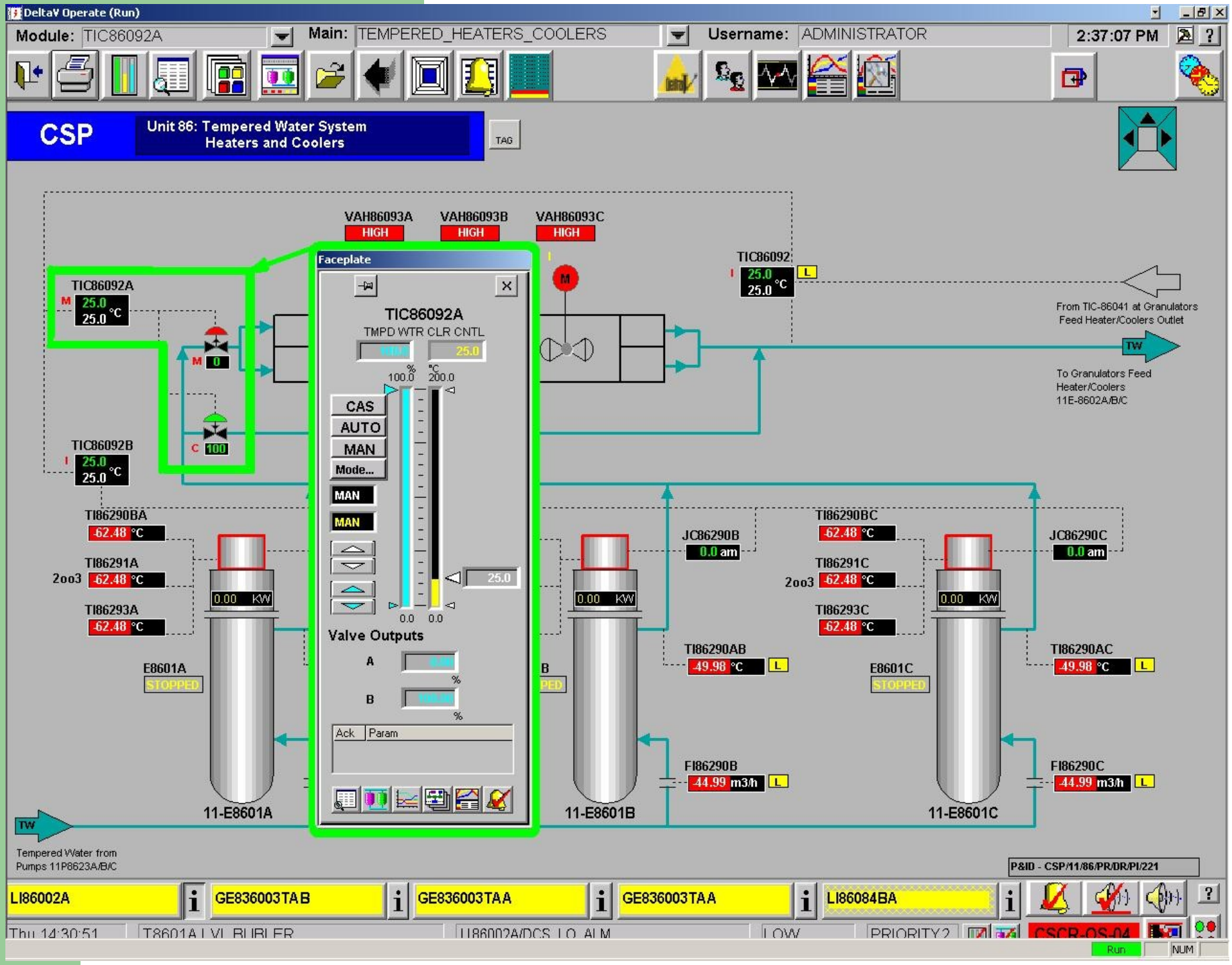

DeltaV Operate (Run)
Module: TI94012
Main: POTABLE_WATER_OVERVIEW
Username: ADMINISTRATOR
5:59:24 AM
CSP
Unit 94:Potable Water System Overview
TAG
SIC94003A
0.0 %
From Sodium Hypochlorite Feed System
P9403A
SIC94003B
PIC94005
0.00
0.00 bar
Detail
TI94012
T9402 TMP
Limits
Alarms
Priority
Enab
Supp
Hi Hi Lim 0
Hi Lim 38
Lo Lim 10
Lo Lo Lim 0
Alm Hysteresis 0.5 %
Low Cutoff 0
Hi Hi PRIORITY1
Hi PRIORITY2
Lo PRIORITY2
Lo Lo PRIORITY1
PV Bad PRIORITY1
Field Value 22.7
Diagnostics
MERROR
MSTATUS
BLOCK_ERR
Module OK
Tuning
PV Filter TC 0.0 s
Linearization
Direct
Faceplate
TI94012
T9402 TMP
23
°C
100
0
Ack
Param
TI94012
23 °C
P94
P94
To Pump Seal
To Control Room
SW
From Desalinated Water
SIC94019A
0.0 %
Y9401P1A
SIC94019B
0.0 %
Y9401P1B
11-Y9401-XM1
LI94018
24.99 %
CO2
11-Y9401-T1
YV94018A
P&ID - CSP/11/94/PR/DR/PI/301B,2A/B,3
PI95025
AI95030B
VI95007A
VI95006A
VI95005A
Sun 14:57:51
IA DISTRIBUTION HDR PR
PI95025/LO_LO_ALM
LOLO
PRIORITY 1
1116-OS-01
Run
NUM

Fieldbus Devices

Control Valve (PCV, TCV, LCV, FCV, etc)

- Control Station:
  - Open Control Faceplate
  - Set PID mode to Auto (set mode to cascade for cascade PID control)
  - Set PID set point to 50%
- Field:
  - force input to 0%, 50%, 100% of span (Inject signal – pressure, temperature, etc. whichever is applicable. Or simulate PV using AMS)
  - verify correct Valve action (control action, fail action)

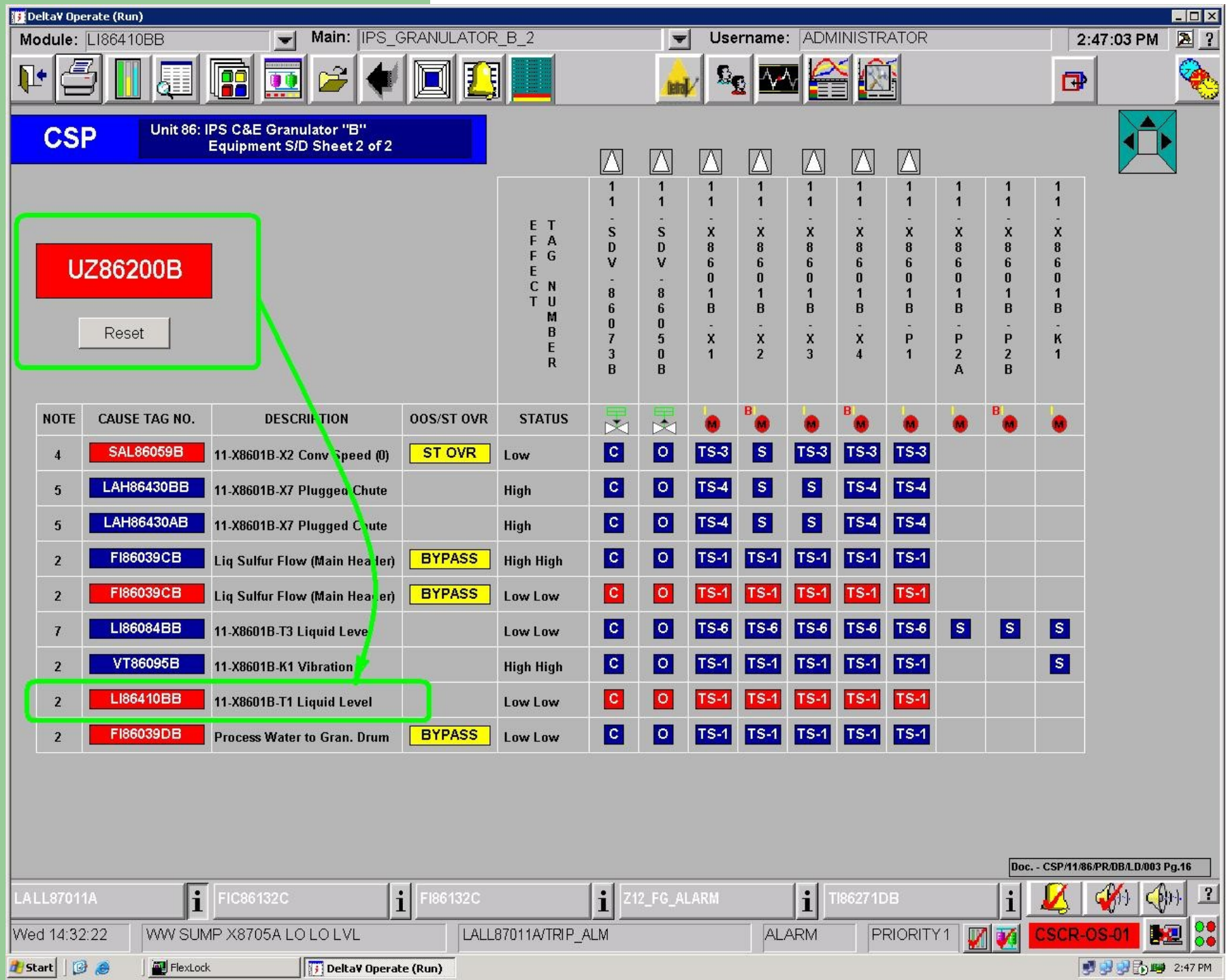

DeltaV Operate (Run)
Module: TV86092AB
Main: TEMPERED_HEATERS_COOLERS
Username: ADMINISTRATOR
2:33:03 PM
CSP
Unit 86: Tempered Water System
Heaters and Coolers
TAG
VAH86093A
HIGH
VAH86093B
HIGH
VAH86093C
HIGH
TIC86092
25.0
25.0 °C
TIC86092A
25.0
25.0 °C
From TIC-86041 at Granulators
Feed Heater/Coolers Outlet
TW
To Granulators Feed
Heater/Coolers
11E-8602A/B/C
A 0
C 100
TIC86092B
25.0
25.0 °C
Faceplate
TV86092AB
E8603 VLV BYPASS
100.0 %
AUTO
CAS
MAN
Mode...
CAS
CAS
0.0
Ack
Param
TI86290BA
-62.48 °C
TI86291A
2oo3
-62.48 °C
TI86293A
-62.48 °C
JC86290A
0.0 am
0.00 KW
E8601A
STOPPED
TI86290A
-49.98 °C
FI86290A
-44.99 m3/h
11-E8601A
JC86290B
0.0 am
TI86290AB
-49.98 °C
FI86290B
-44.99 m3/h
TI86290BC
-62.48 °C
TI86291C
2oo3
-62.48 °C
TI86293C
-62.48 °C
JC86290C
0.0 am
0.00 KW
E8601C
STOPPED
TI86290AC
-49.98 °C
FI86290C
-44.99 m3/h
11-E8601C
TW
Tempered Water from
Pumps 11P8623A/B/C
P&ID - CSP/11/86/PR/DR/PI/221
LI86002A
GE836003TAB
GE836003TAA
GE836003TAA
LI86084BA
Thu 14:30:51
T8601A LVL BUBLER
LOW
PRIORITY2
Run
NUM

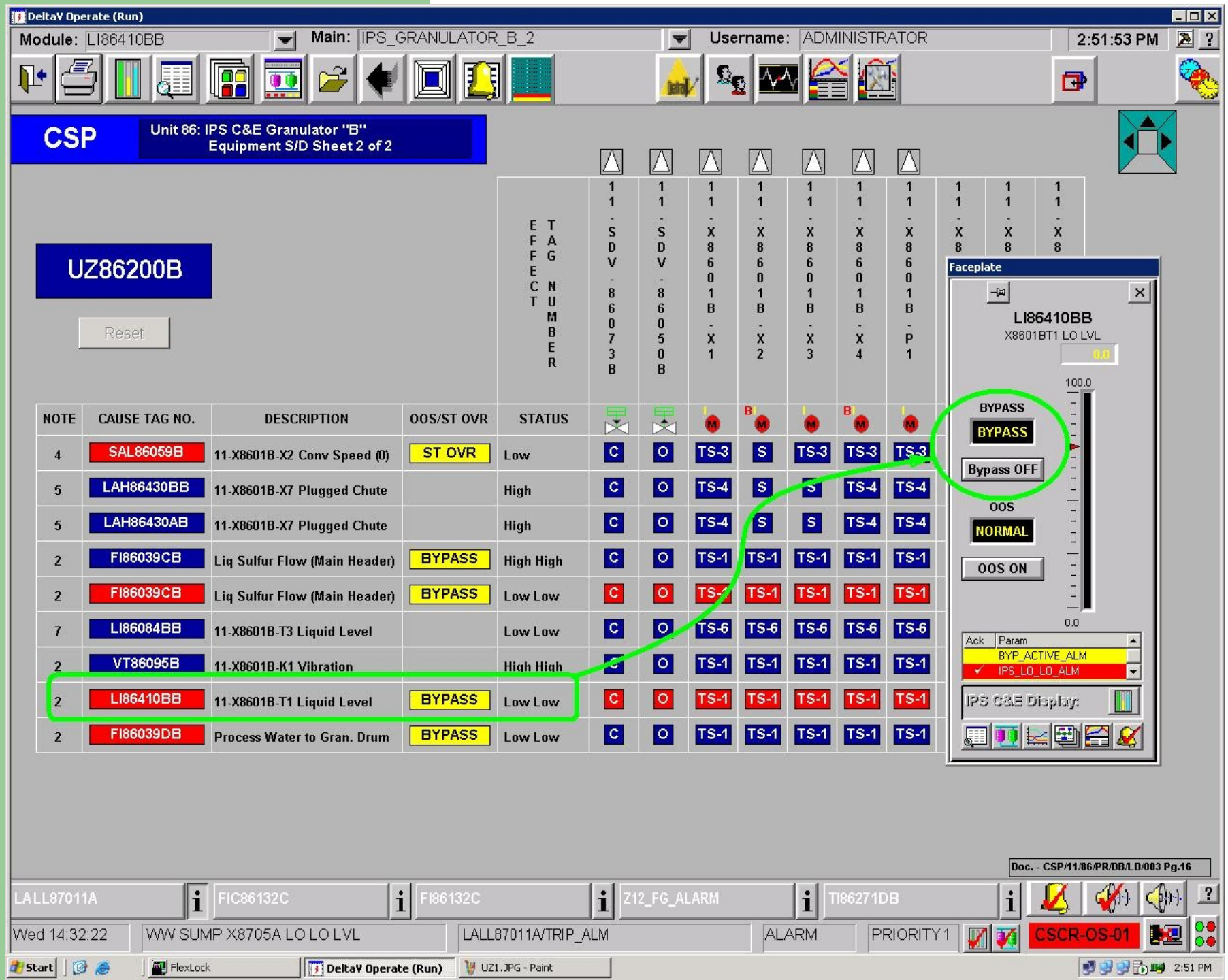

DeltaV Operate (Run)
Module: TIC86092A
Main: TEMPERED_HEATERS_COOLERS
Username: ADMINISTRATOR
2:37:07 PM
CSP
Unit 86: Tempered Water System
Heaters and Coolers
TAG
VAH86093A
HIGH
VAH86093B
HIGH
VAH86093C
HIGH
Faceplate
TIC86092A
TMPD WTR CLR CNTL
CAS
AUTO
MAN
Mode...
MAN
MAN
Valve Outputs
A
B
Ack
Param
TIC86092A
25.0
25.0 °C
TIC86092B
25.0
25.0 °C
TIC86092
25.0
25.0 °C
From TIC-86041 at Granulators
Feed Heater/Coolers Outlet
To Granulators Feed
Heater/Coolers
11E-8602A/B/C
TI86290BA
-62.48 °C
TI86291A
2oo3
-62.48 °C
TI86293A
-62.48 °C
TI86290BB
-62.48 °C
TI86291B
2oo3
-62.48 °C
TI86293B
-62.48 °C
TI86290BC
-62.48 °C
TI86291C
2oo3
-62.48 °C
TI86293C
-62.48 °C
JC86290B
0.0 am
JC86290C
0.0 am
TI86290AB
-49.98 °C
TI86290AC
-49.98 °C
FI86290B
-44.99 m3/h
FI86290C
-44.99 m3/h
E8601A
STOPPED
E8601C
STOPPED
0.00 KW
11-E8601A
11-E8601B
11-E8601C
TW
Tempered Water from
Pumps 11P8623A/B/C
P&ID - CSP/11/86/PR/DR/PI/221
LI86002A
GE836003TAB
GE836003TAA
GE836003TAA
LI86084BA
Thu 14:30:51
T8601A LVL BUBLER
LOW
PRIORITY2
CSCR-OS-04
Run
NUM

IPS signal

Analog Input (AI - Conventional, HART)

- Open IPS C&E display
- Click “Reset” button to clear inactive alarm
- Field:
  - force input to 0%, 50%, 100% of span (Inject signal – pressure, temperature, etc. whichever is applicable. Or simulate PV using AMS)
- Control Station:
  - verify correct reading in DeltaV operate (Control graphics / Faceplate)
  - verify alarm (HH, Hi, Lo, LL) Settings (Detailed Faceplate)


DeltaV Operate (Run)
Module: LI86410BB
Main: IPS_GRANULATOR_B_2
Username: ADMINISTRATOR
2:47:03 PM
CSP
Unit 86: IPS C&E Granulator "B"
Equipment S/D Sheet 2 of 2
UZ86200B
Reset
EFFECT TAG NUMBER
11-SDV-86073B
11-SDV-86050B
11-X8601B-X1
11-X8601B-X2
11-X8601B-X3
11-X8601B-X4
11-X8601B-P1
11-X8601B-P2A
11-X8601B-P2B
11-X8601B-K1
NOTE
CAUSE TAG NO.
DESCRIPTION
OOS/ST OVR
STATUS
4 SAL86059B 11-X8601B-X2 Conv Speed (0) ST OVR Low C O TS-3 S TS-3 TS-3 TS-3
5 LAH86430BB 11-X8601B-X7 Plugged Chute High C O TS-4 S S TS-4 TS-4
5 LAH86430AB 11-X8601B-X7 Plugged Chute High C O TS-4 S S TS-4 TS-4
2 FI86039CB Liq Sulfur Flow (Main Heater) BYPASS High High C O TS-1 TS-1 TS-1 TS-1 TS-1
2 FI86039CB Liq Sulfur Flow (Main Heater) BYPASS Low Low C O TS-1 TS-1 TS-1 TS-1 TS-1
7 LI86084BB 11-X8601B-T3 Liquid Level Low Low C O TS-6 TS-6 TS-6 TS-6 TS-6 S S S
2 VT86095B 11-X8601B-K1 Vibration High High C O TS-1 TS-1 TS-1 TS-1 TS-1 S
2 LI86410BB 11-X8601B-T1 Liquid Level Low Low C O TS-1 TS-1 TS-1 TS-1 TS-1
2 FI86039DB Process Water to Gran. Drum BYPASS Low Low C O TS-1 TS-1 TS-1 TS-1 TS-1
Doc. - CSP/11/86/PR/DB/LD/003 Pg.16
LALL87011A
FIC86132C
FI86132C
Z12_FG_ALARM
TI86271DB
Wed 14:32:22
WW SUMP X8705A LO LO LVL
LALL87011A/TRIP_ALM
ALARM
PRIORITY 1
CSCR-OS-01
Start
FlexLock
DeltaV Operate (Run)
2:47 PM


DeltaV Operate (Run)
Module: LI86410BB
Main: IPS_GRANULATOR_B_2
Username: ADMINISTRATOR
2:51:53 PM
CSP
Unit 86: IPS C&E Granulator "B"
Equipment S/D Sheet 2 of 2
UZ86200B
Reset
EFFECT TAG NUMBER
11-SDV-86073B
11-SDV-86050B
11-X8601B-X1
11-X8601B-X2
11-X8601B-X3
11-X8601B-X4
11-X8601B-P1
11-X8
11-X8
11-X8
NOTE
CAUSE TAG NO.
DESCRIPTION
OOS/ST OVR
STATUS
4 SAL86059B 11-X8601B-X2 Conv Speed (0) ST OVR Low C O TS-3 S TS-3 TS-3 TS-3
5 LAH86430BB 11-X8601B-X7 Plugged Chute High C O TS-4 S S TS-4 TS-4
5 LAH86430AB 11-X8601B-X7 Plugged Chute High C O TS-4 S S TS-4 TS-4
2 FI86039CB Liq Sulfur Flow (Main Header) BYPASS High High C O TS-1 TS-1 TS-1 TS-1 TS-1
2 FI86039CB Liq Sulfur Flow (Main Header) BYPASS Low Low C O TS-1 TS-1 TS-1 TS-1 TS-1
7 LI86084BB 11-X8601B-T3 Liquid Level Low Low C O TS-6 TS-6 TS-6 TS-6 TS-6
2 VT86095B 11-X8601B-K1 Vibration High High C O TS-1 TS-1 TS-1 TS-1 TS-1
2 LI86410BB 11-X8601B-T1 Liquid Level BYPASS Low Low C O TS-1 TS-1 TS-1 TS-1 TS-1
2 FI86039DB Process Water to Gran. Drum BYPASS Low Low C O TS-1 TS-1 TS-1 TS-1 TS-1
Faceplate
LI86410BB
X8601BT1 LO LVL
0.0
100.0
BYPASS
BYPASS
Bypass OFF
OOS
NORMAL
OOS ON
0.0
Ack
Param
BYP_ACTIVE_ALM
IPS_LO_LO_ALM
IPS C&E Display:
Doc. - CSP/11/86/PR/DB/LD/003 Pg.16
LALL87011A
FIC86132C
FI86132C
Z12_FG_ALARM
TI86271DB
Wed 14:32:22
WW SUMP X8705A LO LO LVL
LALL87011A/TRIP_ALM
ALARM
PRIORITY 1
CSCR-OS-01
Start
FlexLock
DeltaV Operate (Run)
UZ1.JPG - Paint
2:51 PM

IPS signal

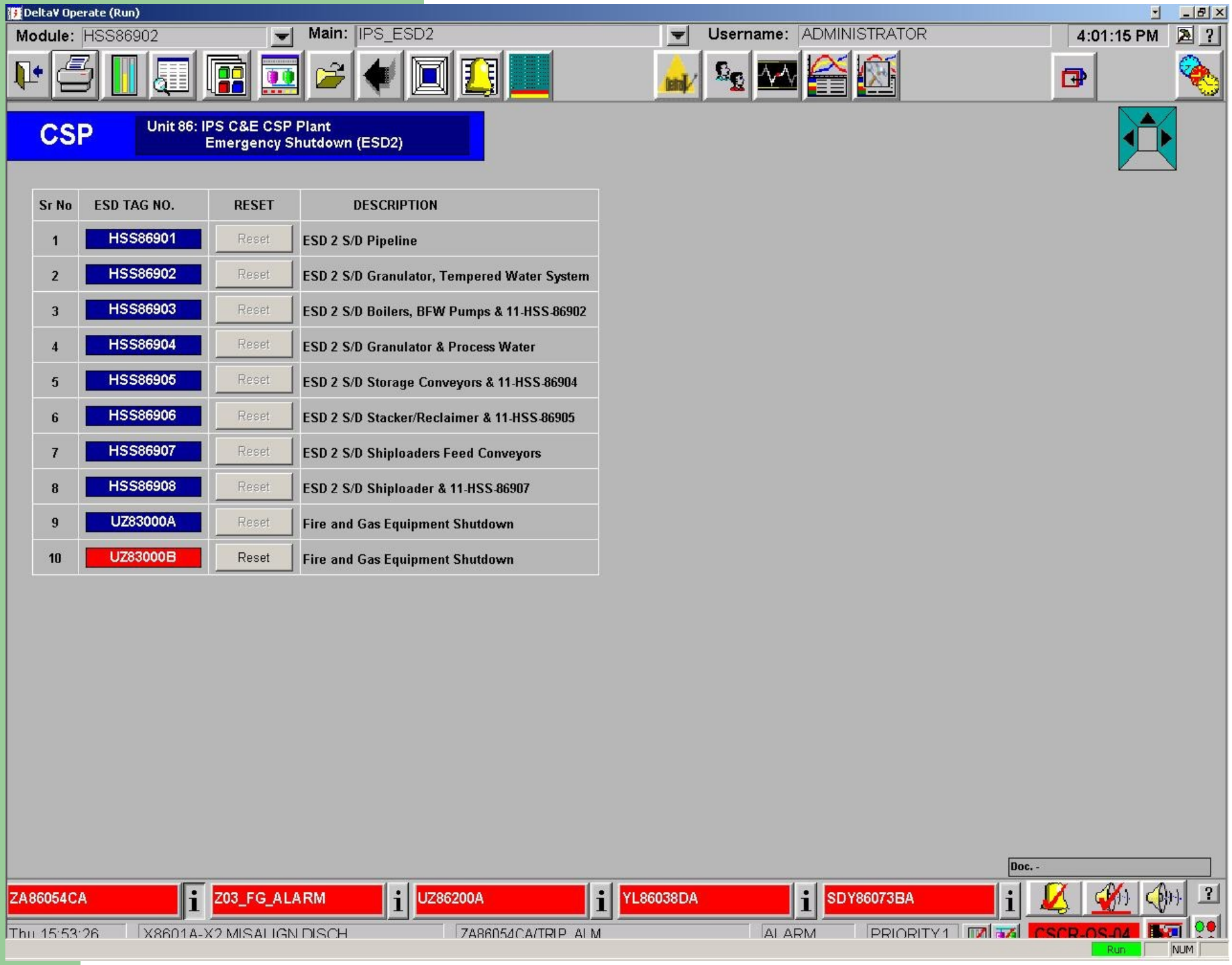
Digital input

- Open IPS C&E display
- Click “Reset” button to clear inactive alarm
- Field:
  - activate field device if applicable.
  - (switches –level, pressure, temperature, flow, Emergency Stop, etc.)
- Control Station:
  - verify correct indication in DeltaV operate (Control graphics / Faceplate) if applicable.

IPS signal

Digital output

- Open IPS C&E display
- Make the related input in the UZ logic healthy.
- Click “Reset” button to clear inactive alarm
- Field:
  - simulate any shutdown signal from the field. (HH, LL, 2oo3 LL, etc.)
  - Verify shutdown signal becomes active (SDV deactivated, Breakers off)
- Control Station:
  - Verify UZ display shows shutdown


DeltaV Operate (Run)
Module: HSS86902
Main: IPS_ESD2
Username: ADMINISTRATOR
4:01:15 PM
CSP
Unit 86: IPS C&E CSP Plant
Emergency Shutdown (ESD2)
Sr No | ESD TAG NO. | RESET | DESCRIPTION
1 | HSS86901 | Reset | ESD 2 S/D Pipeline
2 | HSS86902 | Reset | ESD 2 S/D Granulator, Tempered Water System
3 | HSS86903 | Reset | ESD 2 S/D Boilers, BFW Pumps & 11-HSS-86902
4 | HSS86904 | Reset | ESD 2 S/D Granulator & Process Water
5 | HSS86905 | Reset | ESD 2 S/D Storage Conveyors & 11-HSS-86904
6 | HSS86906 | Reset | ESD 2 S/D Stacker/Reclaimer & 11-HSS-86905
7 | HSS86907 | Reset | ESD 2 S/D Shiploaders Feed Conveyors
8 | HSS86908 | Reset | ESD 2 S/D Shiploader & 11-HSS-86907
9 | UZ83000A | Reset | Fire and Gas Equipment Shutdown
10 | UZ83000B | Reset | Fire and Gas Equipment Shutdown
Doc. -
ZA86054CA
Z03_FG_ALARM
UZ86200A
YL86038DA
SDY86073BA
Thu 15:53:26
X8601A-X2 MISALIGN DISCH
ZA86054CA/TRIP ALM
ALARM
PRIORITY 1
CSCR-OS-04
Run
NUM